REQUIREMENTS SPECIFICATION REPORT

PROJECT: Traduce Señas

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NATIONAL LEARNING SERVICE – SENA
Software Analysis and Development

2026

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Introduction

This report presents the requirements specification for the Traduce Señas software, developed as a training project for the Technologist in Software Analysis and Development program at the National Learning Service (SENA). Its purpose is to document, in a verifiable and unambiguous way, what the system must do and at what quality level, so that it serves as an agreement among the development team, the instructor, and future users.

This document follows the structure recommended by the IEEE 830-1998 standard (Institute of Electrical and Electronics Engineers [IEEE], 1998): it first presents the project context, then describes the product in general terms, and finally details the functional requirements, nonfunctional requirements, and use cases. The requirements described here are the basis for the analysis, design, testing, and implementation documents that accompany this deliverable.

General project information

Problem statement

Deaf people and people with hearing disabilities face persistent barriers when communicating with the hearing population. In Colombia, the number of certified LSC interpreters is far below demand, and availability is even lower in rural areas, public entities, and unplanned everyday situations (Instituto Nacional para Sordos [INSOR], 2020). As a result, deaf people lose autonomy, and their participation in education, employment, and health services is reduced.

In response to this situation, there is a need for a low-cost technological solution, accessible on a smartphone, that translates signs into text or speech without requiring an interpreter in everyday interactions.

Justification

The project promotes social inclusion and equal opportunities for deaf people, in line with Ley 982 de 2005, which recognizes LSC and requires the elimination of communication barriers. It also leverages mature computer vision technologies, such as MediaPipe Hands (Zhang et al., 2020), which make it possible to detect the hand in real time from the camera of a common device, making development feasible for a small team.

General objective

Design and implement a mobile and web application that translates Colombian Sign Language into text or audio in real time, using computer vision and artificial intelligence, to reduce the communication barriers faced by deaf people in everyday settings.

Specific objectives

1. Build a tool that recognizes LSC signs and delivers the translation immediately.
2. Convert the recognized signs into written or spoken words that any hearing person can understand.
3. Enable communication between deaf and hearing people in situations such as the classroom, a store, or a service counter.
4. Validate the tool in real-world scenarios to confirm that it is accurate and useful.
5. Contribute to a more inclusive society in which hearing ability does not determine who can communicate.

Scope

The system comprises an application for Android, iOS, and web browsers that recognizes signs through the camera and delivers the translation as text or audio. It includes account management with roles and permissions, translation history, sign language alphabet lookup (with 3D models), visual notifications, an achievements module, usage statistics for administrators, sample capture and recognition training, an audit trail, error logging, and a configurable password policy.

The following are out of scope for this version: reverse translation (from text or speech to signs), full coverage of the LSC vocabulary, and offline use without an internet connection.

Personnel involved

Table 1
Project team

Name	Role	Responsibility	Contact
Juan Mauricio Suaza Solórzano	Technical lead, Scrum Master, analyst, and developer	Analysis, design, backend programming (lexicon and recognition), and coordination	suazasolorzano@gmail.com
Luis Alejandro Duarte Aldana	Analyst and developer	Analysis, interface design, and programming	aleosea777@gmail.com
Juan Camilo Fierro Esquivel	Analyst and developer	Analysis, product documentation, and programming	fierroesquiveljuancamilo@gmail.com

Note. Apprentices in the Technologist in Software Analysis and Development program at SENA.

Definitions, acronyms, and abbreviations

Table 2
Project glossary

Term	Definition
LSC	Colombian Sign Language, officially recognized by Ley 982 de 2005.
API Gateway	Single entry point to the backend: routes requests, validates the JWT, and limits the request rate.
Microservice	Small, independent service with its own database that handles a single business domain.
JWT	JSON Web Token. Signed credential that identifies the user and their roles in each request.
MediaPipe	Google library that detects the hand in the image and returns 21 anatomical points (landmarks).
Landmark	Each of the 21 hand points, with x, y, z coordinates.
DTW	Dynamic Time Warping. Algorithm that compares two sequences even when they have different speeds.
Template	Saved example of a sign against which the camera input is compared.
Threshold	Maximum accepted distance for displaying a sign as recognized (currently 1.1).
RabbitMQ	Message broker that carries domain events between microservices.
Domain event	Message that reports that something occurred, for example "TranslationProduced".
RBAC	Role-based access control (USER, ADMIN, LINGUIST).
Expo	Platform that allows compiling the same React Native application for Android, iOS, and web.
RF / RNF	Functional requirement / nonfunctional requirement.
CU	Use case.

Overall description

Product perspective

Traduce Señas is a new, independent product. It is designed to be educational, accessible, and easy to use, so that anyone, even without prior knowledge, can interact with the deaf community.

Technically, the system consists of a client built with React Native and Expo (running on Android, iOS, and web browsers) and a microservices backend in Node.js, Express, and TypeScript: an API Gateway and five domain services (iam, recognition, lexicon, analytics, and profile), each with its own PostgreSQL database, MongoDB for training samples, and RabbitMQ for asynchronous event-based communication.

User characteristics

Table 3
User profiles

Profile	Training	Main activities
Deaf user or user with hearing loss	No technical training required	Performs signs in front of the camera, reviews their history and the alphabet, and configures theme and language
Hearing user	No LSC knowledge required	Reads or listens to the translation and learns basic signs with the alphabet
Administrator	Technologist in Software Analysis and Development	Monitors recognition accuracy, manages the lexicon and users, captures samples, reviews statistics and audit logs

Constraints

- **Limited vocabulary:** recognition initially covers the alphabet and a reduced set of words; it does not cover the full LSC.
- **Connectivity:** registration, sign-in, history, and statistics require an internet connection.
- **Hardware:** the device must have a camera and the capacity to process video in real time.
- **Personal data:** the system must comply with Ley 1581 de 2012 and Decreto 1377 de 2013, especially for video samples and hand points.
- **Team and timeline:** three apprentices working part-time and following the SENA

academic calendar.

Assumptions and dependencies

- Deaf and hearing people are willing to use the application to communicate.
- Institutional support is available for pilot testing in real-world settings.
- Public LSC data, such as the LSC-54 dataset (Mora-Zarate et al., 2025), is available to build the recognition templates.
- Users have access to a smartphone or a computer with a camera.
- The system depends on MediaPipe Tasks Vision (hand and pose detection), a custom TensorFlow.js model trained on 54 signs, and an SMTP mail server. It does not use Firebase: notifications are delivered in the app and by email (ADR-007).

Specific requirements

Common interface requirements

User interfaces

The interface is inclusive, intuitive, and accessible: large buttons (minimum 48 × 48 dp), clear iconography, a minimum contrast of 4.5:1 in light and dark themes, and no information conveyed by sound alone. It contains a live video area for recognition, a translation panel, accessibility controls, and an alphabet section.

Hardware interfaces

- Device camera with a minimum resolution of 720p.
- Speaker or audio output for voice mode.
- Stable internet connection.
- Processor capable of running computer vision inference in real time.

Software interfaces

- Android 7.0 (API 24) or later.
- iOS 15.1 or later.
- Up-to-date desktop browsers with WebAssembly and camera access support (Chrome, Edge, Firefox).

- Server with Docker and Docker Compose for the backend.

Communication interfaces

- HTTPS for all client-server connections.
- REST API with data exchange in JSON format.
- JWT authentication in the "Authorization: Bearer" header.
- AMQP messaging (RabbitMQ) between microservices, only within the internal network.

Functional requirements

Table 4 summarizes the eleven functional requirements of the system and their specifications (ERF). The detailed catalog follows, dividing each requirement into atomic, verifiable functions numbered by module.

Table 4
System functional requirements

Code	Requirement	Specifications	Priority
RF1	User management	ERF1.1 Registration · ERF1.2 Sign-in (own account, Google, Facebook) · ERF1.3 Password recovery	High
RF2	Site navigation	ERF2.1 Navigation bar visible on all devices	High
RF3	Sign language translation	ERF3.1 Translation mode activation · ERF3.2 Conversion of signs to text/audio	High
RF4	Visual notifications	ERF4.1 On-screen notifications	Medium
RF5	Profile management	ERF5.1 View profile · ERF5.2 Light/dark theme · ERF5.3 Change password · ERF5.4 Language · ERF5.5 Recent translations	High
RF6	Usage statistics (administrator only)	ERF6.1 Access to statistics	Medium
RF7	Sign language alphabet	ERF7.1 Alphabet display	High
RF8	Gamification	ERF8.1 Achievement catalog and progress · ERF8.2 Achievement unlock and notice · ERF8.3 Daily challenges and challenge-point medals	Low
RF9	Internal audit	ERF9.1 Audit trail query	Medium
RF10	Error handling	ERF10.1 Error classification and query	Medium
RF11	Password policy	ERF11.1 Policy configuration	Medium

Note. RF8 (gamification) is implemented, including daily challenges and medals according to ADR-009 and ADR-010. RF9 and RF10 are not implemented yet. RF11 only has password validation with fixed rules (FR-906); the configurable policy does not exist yet.

Identity module (iam-service)

Table 5
Detailed requirements — Identity (iam-service)

ID	Description	Source	Priority
FR-001	Register a user with name, email, a valid password, and acceptance of the terms with date and version, without birth date or username; the account is created as INACTIVE until the email is verified	RF1	Must
FR-002	Reject registration unless the terms are accepted, and store the accepted date and version	RF1	Must
FR-003	Verify the email with a 6-digit code	RF1	Must
FR-004	Store codes only as hashes, with expiration, single use, and an attempt counter	RF1	Must
FR-005	Authenticate with email and password	RF1	Must
FR-006	Temporarily lock the account after failed attempts (1, 5, and 10 minutes)	RF1	Must
FR-008	Registration and sign-in with Google and Facebook	RF1	Must
FR-010	Recover the password with a single-use code that expires	RF1	Must
FR-012	Withdrawn. Formerly: rotate refresh tokens. No refresh token exists: the session is a JWT that lasts one day	RF1	—
FR-014	Delete the account, confirmed with a code sent to the email: the user row is deleted, and the iam.UserDeleted event makes the other services erase or unlink their copies (Ley 1581 de 2012)	RF1, RF5	Should
FR-015	The administrator lists, locks, restores, and deactivates accounts	RF1, RF6	Should

Note. MoSCoW priority (Must, Should, Could). The complete catalog contains 88 functional requirements, 85 active and 3 withdrawn (04-requirements/functional.md); these tables show the representative ones for each module.

Authorization module (iam-service)

Table 6
Detailed requirements — Authorization (iam-service)

ID	Description	Source	Priority
FR-101	Role-based access control for USER, ADMIN, and LINGUIST	RF6	Must
FR-102	Permissions are assigned to roles, never directly to users	RF6	Must
FR-103	Deny administration functions to anyone without the permission	RF6	Must
FR-104	Do not allow deleting a role that is assigned to a user	RF6	Must

Profile and notifications module (profile-service)

Table 7
Detailed requirements — Profile and notifications (profile-service)

ID	Description	Source	Priority
FR-201	View and edit profile data	RF5	Must
FR-202	Change the password in compliance with the current policy	RF5	Must
FR-203	Choose a light or dark theme	RF5	Should

ID	Description	Source	Priority
FR-204	Choose the interface language (Spanish, English, French, and Portuguese)	RF5	Should
FR-205	Enable or disable all notifications with a single toggle	RF4, RF5	Must
FR-502	Deliver notifications only through visual channels: in-app and email (no push, ADR-007)	RF4	Must
FR-503	Withdrawn (ADR-007). Formerly: register devices for push notifications. Push is replaced by email, and no device table exists	RF4	—
FR-507	Mark a notification as read	RF4	Must
FR-509	Withdrawn (ADR-007). Formerly: every push notification must have a target device. No push notification is created	RF4	—
FR-510	Send administrators with notifications enabled the weekly and monthly usage reports and the weekly model reminder (MODEL_REMINDER), as an in-app notice and by email, with a button to the admin panel	RF4, RF6	Should

Lexicon module (lexicon-service)

Table 8

Detailed requirements — Lexicon (lexicon-service)

ID	Description	Source	Priority
FR-301	Present the sign catalog organized by category	RF7	Must
FR-302	Search signs by name	RF7	Must
FR-303	Show the details of a sign with its multimedia resources	RF7	Must
FR-304	Present the alphabet (A–Z and Ñ) without repeated letters	RF7	Must
FR-305	Each sign has a stable code that does not change when published	RF7	Must
FR-307	The administrator creates, edits, publishes, and withdraws signs	RF7	Must
FR-310	Only ACTIVE signs are visible to the end user	RF7	Must

Translation and AI module (recognition-service)

Table 9

Detailed requirements — Translation and AI (recognition-service)

ID	Description	Source	Priority
FR-401	Capture signs with the camera and translate them into text in real time	RF3	Must
FR-402	Provide audio output, always accompanied by visible text	RF3	Should
FR-403	Record the confidence and the source (engine) of each translation	RF3	Must
FR-405	Visually warn when a sign is not recognized, instead of showing an incorrect translation	RF3	Must
FR-406	View one's own translation history, from most recent to oldest	RF3, RF5	Must
FR-407	Delete history entries (logical deletion)	RF5	Should
FR-409	Show the history grouped by sessions, with one card per session	RF3, RF5	Should
FR-410	Delete history translations 30 days after they were created	RF5	Should

ID	Description	Source	Priority
FR-411	The administrator adds words without retraining the model: they are saved as word-type templates (30 frames of 128 values) and, on the device, are matched with DTW before the model (ADR-011)	RF3	Should
FR-601	Capture gesture samples grouped by capture session	RF3	Must
FR-602	Do not store any sample without verifiable consent	—	Must
FR-605	Maintain a registry of models with their training metrics	RF3	Must

Usage, achievements, audit, and configuration module

Table 10
Detailed requirements — Usage, achievements, audit, and configuration

ID	Description	Source	Priority
FR-701	Record an event each time the user enters a section	RF6	Must
FR-703	Show the administrator the most visited sections	RF6	Must
FR-706	Statistics are aggregated and require permission	RF6	Must
FR-707	Each web screen has its own URL, so it can be opened, reloaded and shared directly	RF2	Should
FR-801	Maintain an achievements catalog with a target greater than zero	RF8	Could
FR-803	An achievement is unlocked only once per user	RF8	Could
FR-806	Assign each user a daily translation challenge (1, 3, 5 or 8 translations) and show its progress, which updates when translations are saved	RF8	Could
FR-807	When the challenge is completed, store the point reward (5, 10, 15 or 25) and notify the user	RF8, RF4	Could
FR-808	Grant medals based on the total accumulated challenge points	RF8	Could
FR-809	Show an on-screen notice when an achievement is unlocked or a challenge is completed, without repeating it	RF8, RF4	Could
FR-901	Maintain an audit trail with values before and after each change	RF9	Must
FR-904	Classify errors by type (validation, authentication, permission, database, etc.)	RF10	Must
FR-905	Configure the password policy; only one policy may be active	RF11	Should

Nonfunctional requirements

Table 11
Nonfunctional requirements and their acceptance metric

Code	Attribute	Requirement and metric
RNF1	Accessibility	WCAG 2.1 Level AA as the target; no information conveyed by sound alone; contrast 4.5:1; touch targets of 48 dp; notifications use visual channels only
RNF2	Performance	P95 latency below 300 ms on critical services under 50 requests per second; recognition of each sign in under 500 ms; ≥ 15 frames per second during capture; model accuracy ≥ 85% before publishing
RNF3	Availability	Production SLO of 99.5%; without connection, the alphabet and history are read from cache and notifications are deferred

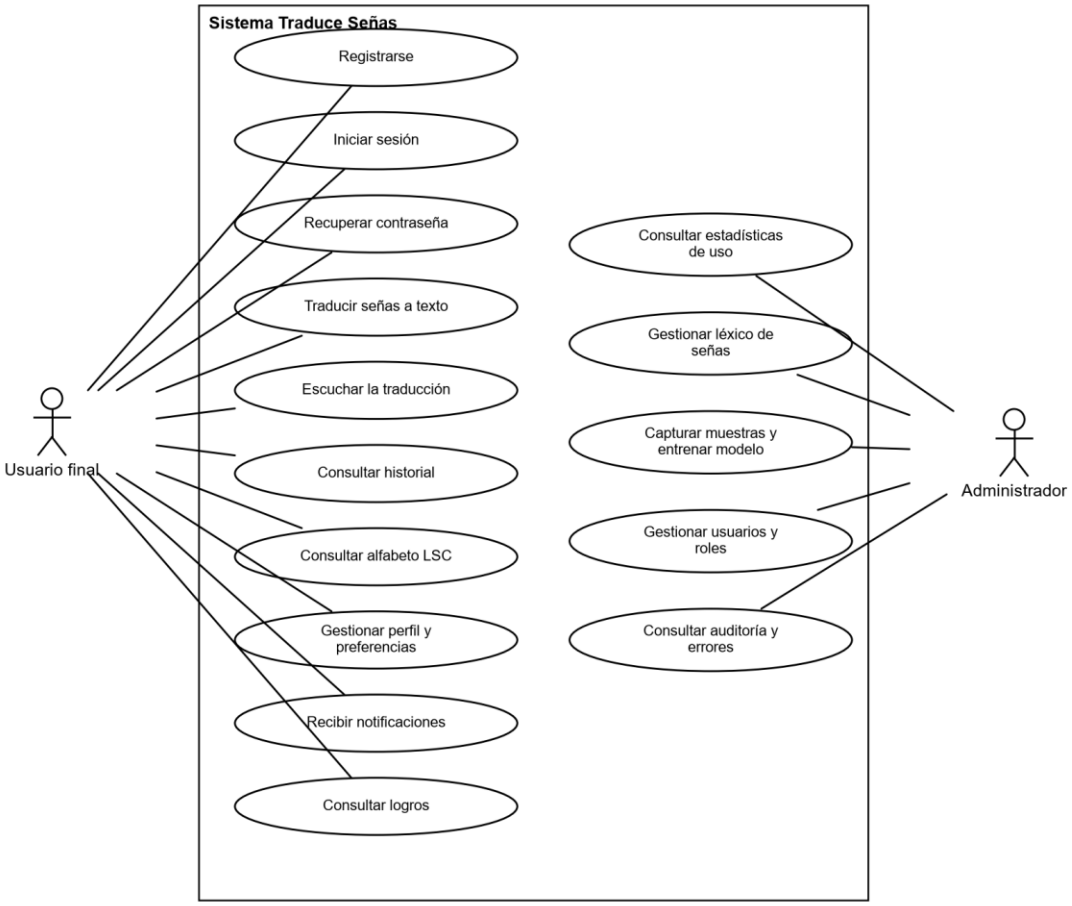
Code	Attribute	Requirement and metric
RNF4	Scalability	500 concurrent users and 5,000 registered; each microservice scales independently; lexicon of up to 5,000 signs without query degradation
RNF5	Security	Passwords hashed with bcrypt (cost 10 today; target 12); mandatory JWT valid for one day, with no refresh token; progressive lockout; limit of 10 requests per minute per IP on sign-in and recovery; HTTPS in production
RNF6	Observability	GET /health endpoint in each service and GET /health/ready for readiness; structured logs with a correlation ID and classified errors
RNF7	Maintainability	Test coverage ≥ 80% of lines (≥ 90% in the domain); cyclomatic complexity ≤ 10 per method; schema changes only through versioned migrations
RNF8	Portability	Backend deployed as a Docker image; PostgreSQL 15 or later and MongoDB 6 or later; configuration only through environment variables; mobile app from Android 7.0 (API 24) and iOS 15.1
RNF9	Usability	First translation in under 3 minutes from opening the app (test with 5 users); registration in 4 screens or fewer; error messages in plain language
RNF10	Disaster recovery	Daily PostgreSQL and MongoDB backups as the target (today, manual dumps), with 30-day retention; RTO under 30 min and RPO under 24 h for PostgreSQL; quarterly restore drill

Note. Accessibility is treated as a primary constraint because the primary users are deaf people (World Wide Web Consortium [W3C], 2018).

Use case requirements

UML use case diagram

Figure 1
General use case diagram



Note. Own elaboration.

Use case characterization

Table 12
Characterization of use case CU-01 Sign in

Element	Description
Actors	End user, Administrator
Requirement	RF1 / ERF1.2
Precondition	The account exists and is ACTIVE
Main flow	1. The user opens the app and selects "Sign in". 2. Enters email and password. 3. The system validates the credentials. 4. The system displays the main dashboard.

Element	Description
Alternative flows	3a. Invalid credentials: displays an error and adds a failed attempt. 3b. Attempts are exceeded: the account is temporarily locked.
Postcondition	The user has an active session with their token

Table 13

Characterization of use case CU-02 Sign up

Element	Description
Actors	End user
Requirement	RF1 / ERF1.1
Precondition	The email is not registered
Main flow	1. The user selects "Sign up". 2. Fills out the form and accepts the terms. 3. The system creates the account and sends a code. 4. The user enters the code and the account becomes active.
Alternative flows	2a. Invalid data or terms not accepted: the system displays the errors. 4a. Expired code: the user requests that it be resent.
Postcondition	ACTIVE account, profile, and preferences created

Table 14

Characterization of use case CU-03 Translate signs to text

Element	Description
Actors	End user
Requirement	RF3 / ERF3.1, ERF3.2
Precondition	Signed-in session and camera permission
Main flow	1. The user opens "Translation" and taps "Start camera". 2. Performs the sign in front of the camera. 3. The system recognizes the sign and displays the text. 4. Optionally plays the voice output. 5. The system saves the translation to the history.
Alternative flows	1a. No camera permission: the system requests it. 3a. Sign not recognized: visual notice; no word is displayed.
Postcondition	The translation is stored in the user's history

Table 15

Characterization of use case CU-04 Consult LSC alphabet

Element	Description
Actors	End user
Requirement	RF7 / ERF7.1
Precondition	None (also available in the demo)
Main flow	1. The user opens "Alphabet". 2. The system displays the letters. 3. The user selects a letter and views its 3D model and description.
Alternative flows	2a. No connection: the letters saved in the cache are displayed.
Postcondition	The user views the selected sign

Table 16
Characterization of use case CU-05 Consult usage statistics

Element	Description
Actors	Administrator
Requirement	RF6 / ERF6.1
Precondition	Session with ADMIN role
Main flow	1. The administrator opens "Statistics". 2. The system displays translations per day, the most visited sections, and the most used signs.
Alternative flows	1a. User without ADMIN role: the system denies access (403).
Postcondition	Aggregated statistics visible

Table 17
Characterization of use case CU-06 Capture samples and train

Element	Description
Actors	Administrator
Requirement	RF3 / FR-601…606
Precondition	Session with ADMIN role and signer consent
Main flow	1. The administrator opens "Training". 2. Selects the sign and captures samples. 3. The system saves the samples. 4. The administrator generates and publishes the templates.
Alternative flows	3a. No consent: the sample is rejected. 2a. Discards the entire capture session.
Postcondition	New templates available for recognition

Traceability

Each functional requirement is linked to the user story that plans it and to the tables that persist it. The complete traceability matrix is maintained in the documentation repository (04-requirements/traceability-matrix.md). Table 18 summarizes the implementation status of the 85 active detailed requirements according to that matrix.

Table 18
Implementation status of the detailed requirements

Status	Detailed requirements
Covered	57
Partial	10
Not implemented	16
Not verified	2
Total	**85**

Conclusions

The specification defines eleven functional requirements, broken down into 88 detailed

requirements (85 active and 3 withdrawn), and ten nonfunctional requirements. Of the 85 active requirements, 57 are covered, 10 are partial, 16 are not implemented, and 2 are not verified. Accessibility and personal data protection are established as primary constraints due to the nature of the users and of the biometric information the system handles.

This document is the baseline for analysis, design, and testing; any change in scope must be recorded here and in the traceability matrix before it is implemented.

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